Chapter 5

Estimation

Introduction

In this chapter, we will learn methods for estimating parameters instead of guessing.

- Method of Moments estimation
- Maximum Likelihood estimation

5.1

Method of Moments Estimators (MMEs)

5.1.1

The idea

Background

Method of moments estimators are constructed by:

- Equating the moments of a population's distribution with the sample moments.
- Solving for the unknown parameters.
- Use as many moments as you need in order to solve for all parameters.

Typically, we start with the lower moments first.

- i.e., $k = 1$ first, then $k = 2$ if necessary, then $k = 3$ if still necessary, etc.

Definition 5.1.1

The k**th population (distribution) moment** of X, for $k = 1, 2, \ldots$, is denoted by μ_k, and defined as
$\mu_k := E[X^k]$.
Note that $\mu_1 = E[X^1] = \mu$.

Definition 5.1.2

The k**th sample moment** of X, for $k = 1, 2, \ldots$, is denoted by M_k, and defined as

$$M_k := \frac{1}{n} \sum_{i=1}^{n} X_i^k.$$

Note that $M_1 = \bar{X}$.

5.1.2

Examples

Example 5.1.1

Let $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} \text{Exponential}(\lambda)$.
Find an MME of λ.

Example 5.1.1 (cont)

Example 5.1.2

Let $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} \mathrm{Gamma}(\alpha, \beta)$.
Find MMEs of α and β.

Example 5.1.2 (cont)

Example 5.1.2 (cont)

Example 5.1.2 (cont)

5.1.3

Properties of MMEs

Pros of MMEs

- Intuitive
- Easy to find, overall.
- Usually consistent.
- Usually asymptotically normal.

Cons of MMEs

- Often biased.
- Not uniquely defined.
- May be outside of the parameter space.
- May not exist.

Example 5.1.1 continued

In Example 5.1.1, we derived the MME $\hat{\lambda} = \bar{X}$ for an $\mathrm{Exponential}\lambda$ population.
What can we say about the convergence of the estimator?

Example 5.1.1 (cont)

Example 5.1.1 (cont)

Example 5.1.2 continued

In Example 5.1.2, we derived the MME estimator

$$\hat{\alpha} = \frac{\bar{X}^2}{\frac{1}{n}\sum_{i=1}^{n} X_i^2 - \bar{X}^2} = \frac{M_1^2}{M_2 - M_1^2}.$$

Example 5.1.2 (cont)

Example 5.1.2 (cont)

Some thoughts

- The WLLN gives us that $M_k \xrightarrow{P} \mu_k$ as long as $\mathrm{var}(X^k)$ is finite.
- Asymptotic normality using the CLT and combining the WLLN with the Delta Method.
- MMEs are not well-defined. We could use the second moments to derive an estimator instead of the first.
- MMEs may take values that fall outside the parameter space.
- MMEs may not exist if moments don't exist (i.e., aren't finite).